Swine Influenza Virus

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Introduction

- The first reports of influenza-like disease in pigs occurred in conjunction with the 1918 human influenza pandemic.
- In the four genus of influenza virus (A-D), only IAVs are of routine clinical significance in swine, though ICV can also infect pigs.
- Infection in pigs are **usually subclinical** but clinical symptoms may get worse when concurrent infection occur (ex PRDC).
- **Human-to-swine transmission of influenza virus is far more frequent than swine-to-human (reverse zoonosis).**

Etiology

Influenza A virus (IAV)

- Genus *Alphainfluenza virus*, family of *Orthomyxoviridae*
- Polymorphic, enveloped & approximately 80-120 nm
- Eight, negative-sense, single-stranded RNA segments

  → **Genetic reassortment** (exchange of RNA segments during co-infection)
- 18 (HA) x 11 (NA) subtypes and numerous genotypes
- Three major IAV of swine: H1N1, H1N2, and H3N2
- **Wide range of host species** (ducks, geese, and waterfowl to humans, equine, dogs, pigs and bats)

Human cases in Taiwan

國內出現第3例自呼吸道疾病病人中分離出H1N2v流感病毒案例，病患已痊癒，疾管署呼籲民眾做好個人健康管理

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疾病管制署今(11)日表示，一名住中部之10多歲青少年流感輕症病患之呼吸道檢體分離出無法次分型之A型流感病毒，經該署進一步檢驗並進行基因定序為H1N2v流感病毒，與我國2021年首例感染H1N2v個案之基因組成類似。經調查個案密切接觸者3名，均未有類流感症狀，經檢驗均未發現H1N2v流感病毒感染。疾管署並邀集農政單位及相關醫學專家召開會議討論，依現有流行病學調查結果研判，本案應屬散發個案，與國際過去案例類似，無證據顯示有人傳人現象，衛生及農政單位亦將持續加強人類、動物與環境之監測。

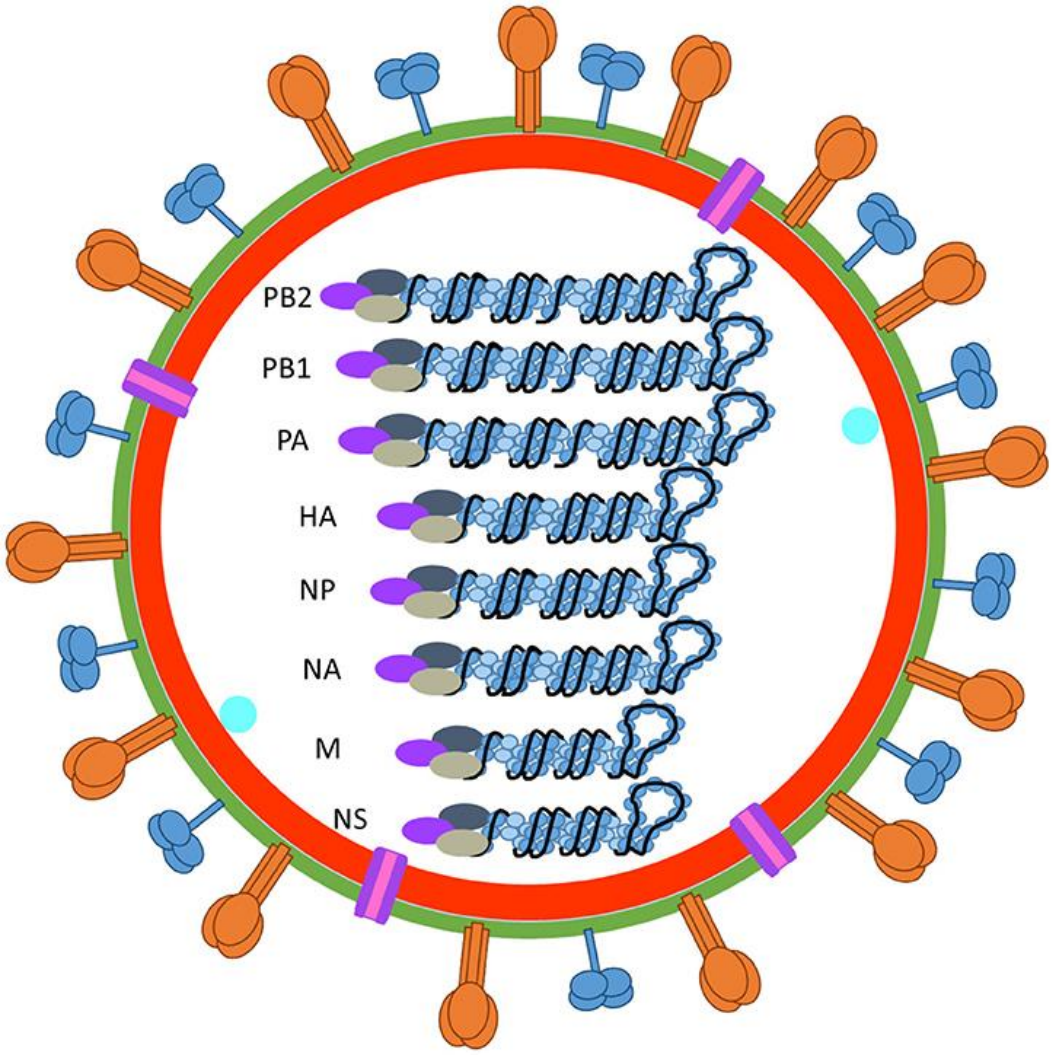
疾管署指出，本案為我國第三例H1N2v流感病毒案例，個案近期無出國旅遊史，疫調後有豬隻接觸史，今(2023)年3月13日出現發燒、鼻塞及肌肉痠痛等症狀並於隔日至診所就醫，流感快篩A型陽性，醫師採檢並開立流感抗病毒藥劑後返家，目前已康復。

疾管署進一步指出，3月28日自該名女性呼吸道檢體培養後分離流感病毒，其它呼吸道感染常見病毒如腺病毒、呼吸道融合病毒、腸病毒及鼻病毒等鑑定後，呈陰性反應。此外，農政單位亦自病例接觸之養豬場豬隻採集檢體檢驗，結果未檢出H1N2流感病毒。針對我國出現第3例確診個案，疾管署將同步透過IHR聯繫窗口通報世界衛生組織。

疾管署說明，我國首例及第2例H1N2v流感病毒案例分別為前(2021)年4月及去(2022)年10月檢出，皆為住在中部之未滿10歲女童，家中皆從事禽畜養殖業，惟前2案與本案並非住在相同縣市，且3案之養殖場位置、飼料與豬隻來源、運送車輛行走路徑等皆不相同，研判3案無關聯性。

H1N2v為存在於豬隻之低病原性流感病毒，偶有人類感染報告案例，且多為輕症。全球自2011年迄今累計至少47例H1N2v流感確診病例，以美洲地區41例為多，部分病例具豬隻接觸史或暴露於受汙染環境，尚無透過食用肉品感染案例。本次由病患所分離之病毒基因序列，與本土過去於豬隻分離之H1N2病毒株較為接近，但不相同，對目前公費流感抗病毒藥物敏感。

Something's Wrong



Protein	Abbreviation
Polymerase basic protein 2	PB2
Polymerase basic protein 1	PB1
Polymerase acidic protein	PA
Hemagglutinin	HA
Nucleoprotein	NP
Neuraminidase	NA
Matrix	M1
	M2
Non-structural protein	NS2
Viral RNA	

Epidemiology

—Why Swine Influenza matters?



https://depositphotos.com/photo/swine-flu-pandemic-spreading-14227901.html

Requirements for Pandemic Influenza virus

1. able to infect humans,
2. able to cause diseases in humans,
3. able to spread efficiently from person to person

Adaptation & fitness

Barriers between Avian & Human IAV

1. **Entry**
2. **Immune evasion**
3. **Replication**

The mixing vessel theory

Receptor–types and distribution

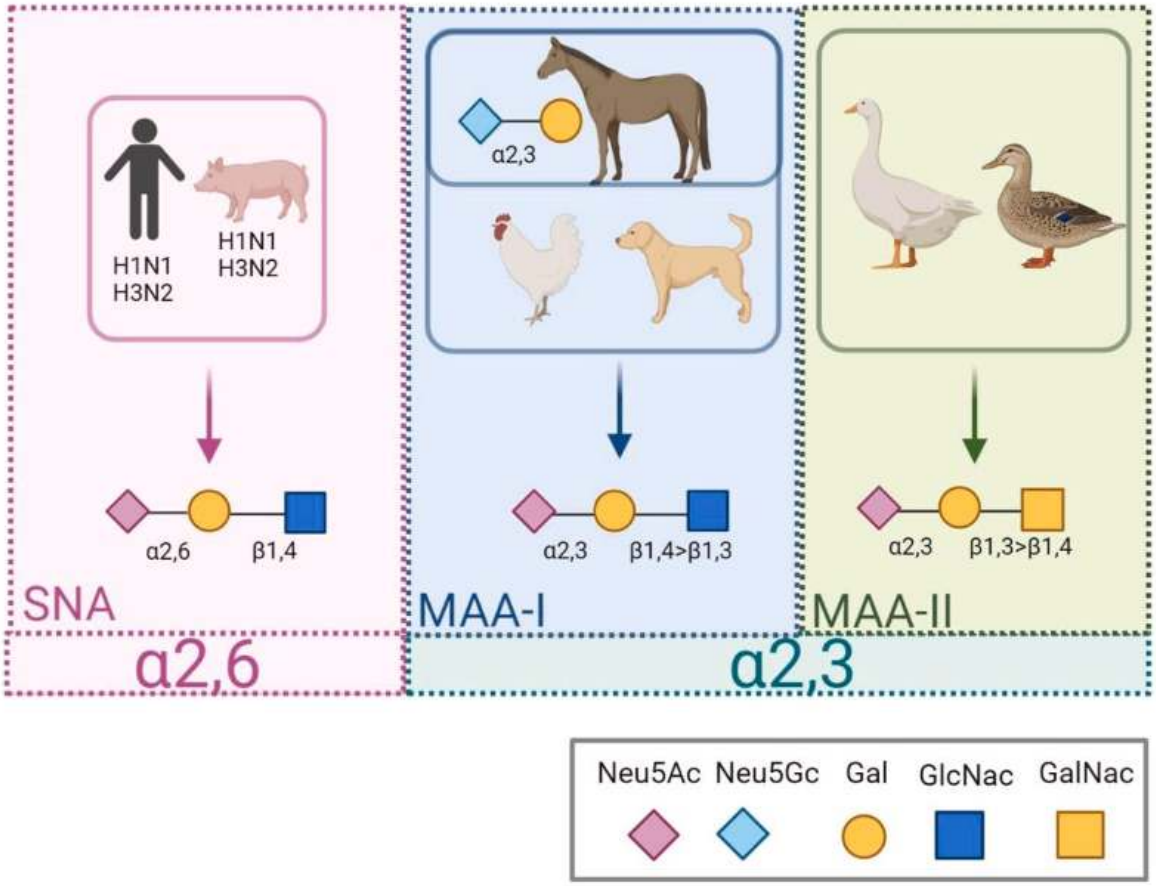


Fig. 1. Illustration of the general receptor binding preferences of different host species of influenza A virus and the main binding affinity of Sambucus Nigra Lectin (SNA), Maackia amurensis Lectin I (MAA-I), and Maackia amurensis Lectin II (MAA-II) lectins. Neu5Ac= N-Acetylneuraminic acid. Neu5Gc= N-Glycolylneuraminic acid. Gal= galactose. GlcNac= N-acetylglucosamine. GalNac= N-acetylgalactosamine. Made in Biorender.com.

IAV receptor: sialic acid

-- Avian IAV HA: α2, 3-linked

-- Human IAV HA: α2, 6-linked

-- Swine IAV HA: α2, 6-linked

-- Two critical aa mutations (190D & 225D) in HA are required for avian-to-human HA switch.

But **sufficient**?

Kristensen et al., 2024

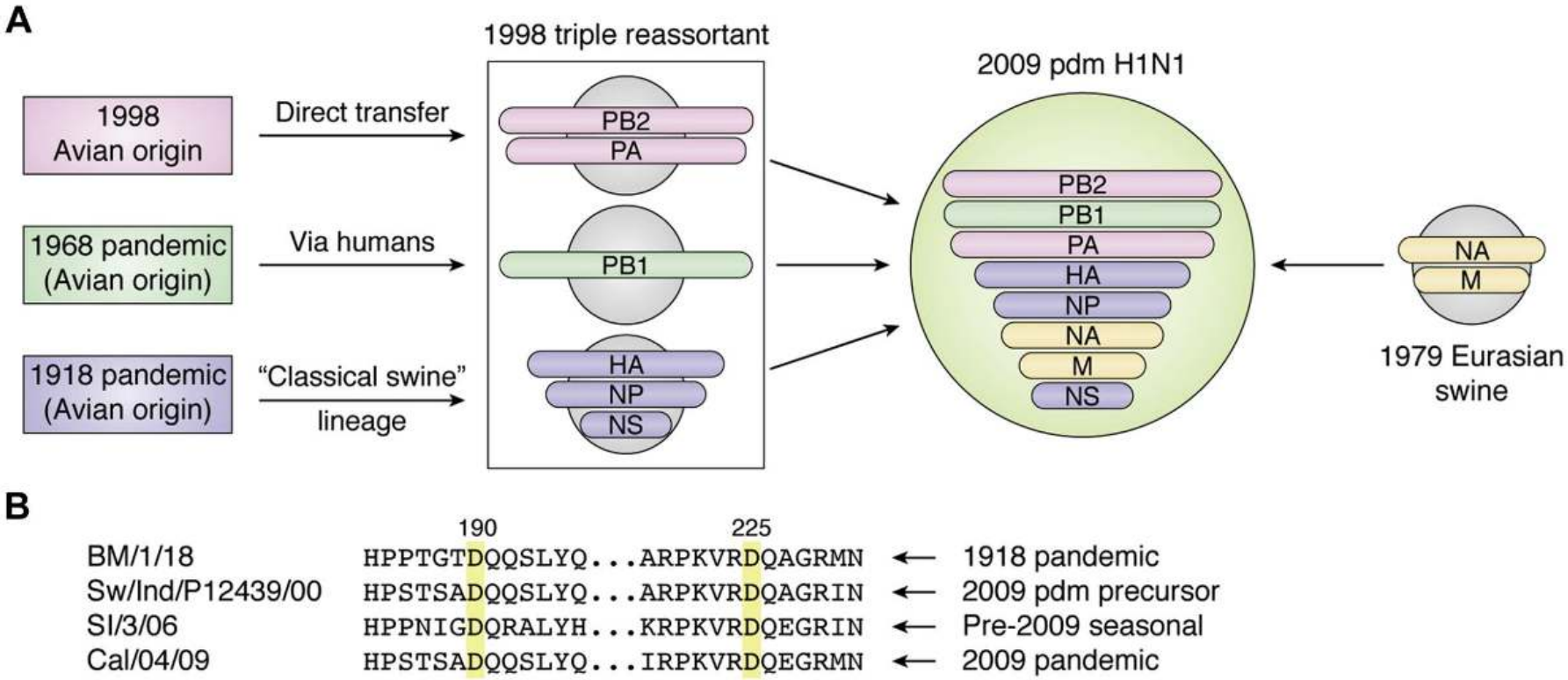
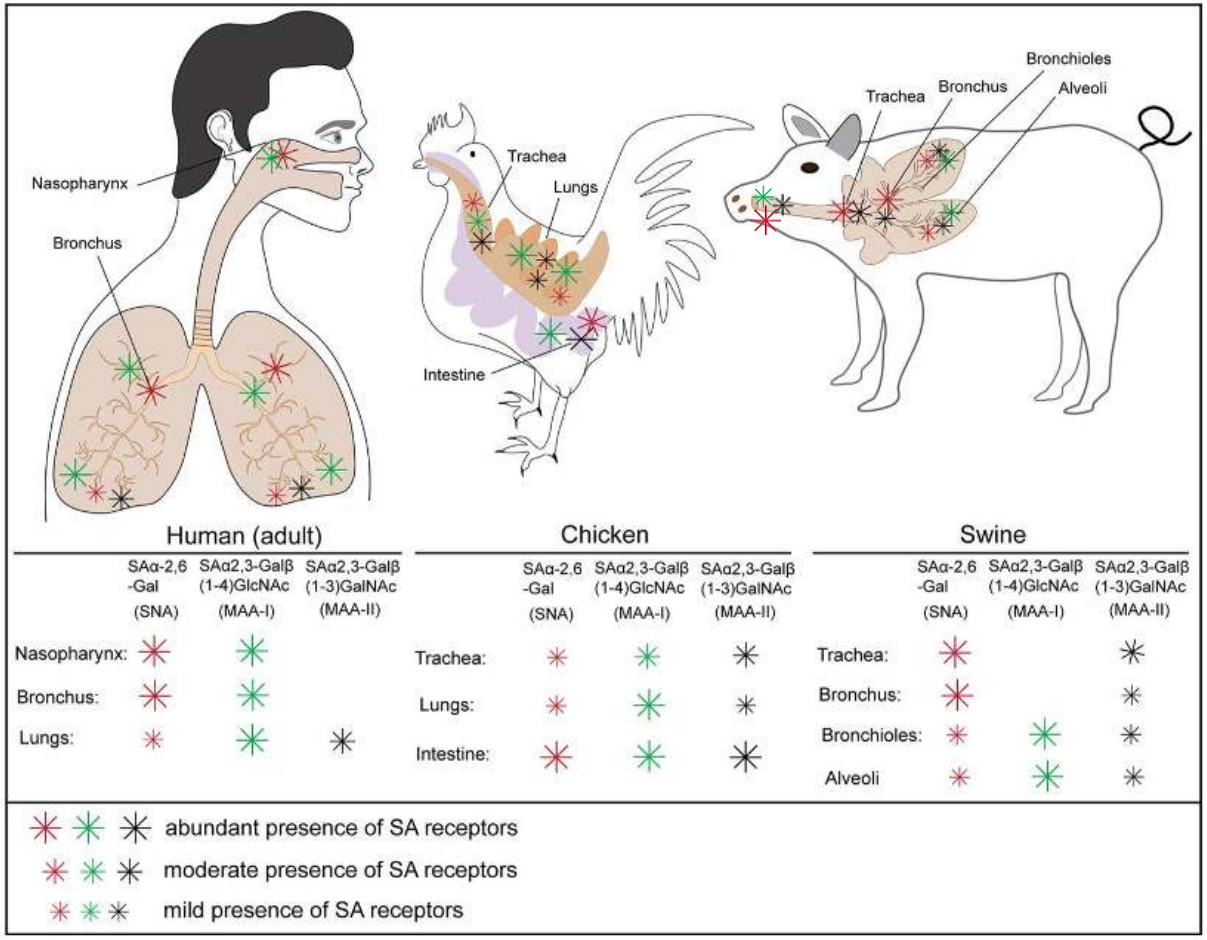


Figure 3. Reassortment leading to emergence of the 2009 pandemic H1N1 virus. *A*, cartoon schematic of various swine lineages contributing gene segments to the novel virus (adapted from data in Garten *et al.* (2009) (24)). *B*, conservation of key humanizing receptor-binding variants in example viruses from the original 1918 pandemic, through the classical swine H1N1 lineage, and within both pre- and post-2009 human H1N1 strains. Strain abbreviations are: A/Brevig Mission/1/1918, A/swine/Indiana/P12439/2000 (nearest identified swine HA gene precursor to 2009 human pandemic strains), A/Solomon Islands/3/2006 (pre-2009 seasonal), and A/California/04/2009 (2009 pandemic).

Thompson & Paulson, 2021

The mixing vessel theory

Receptor–types and distribution

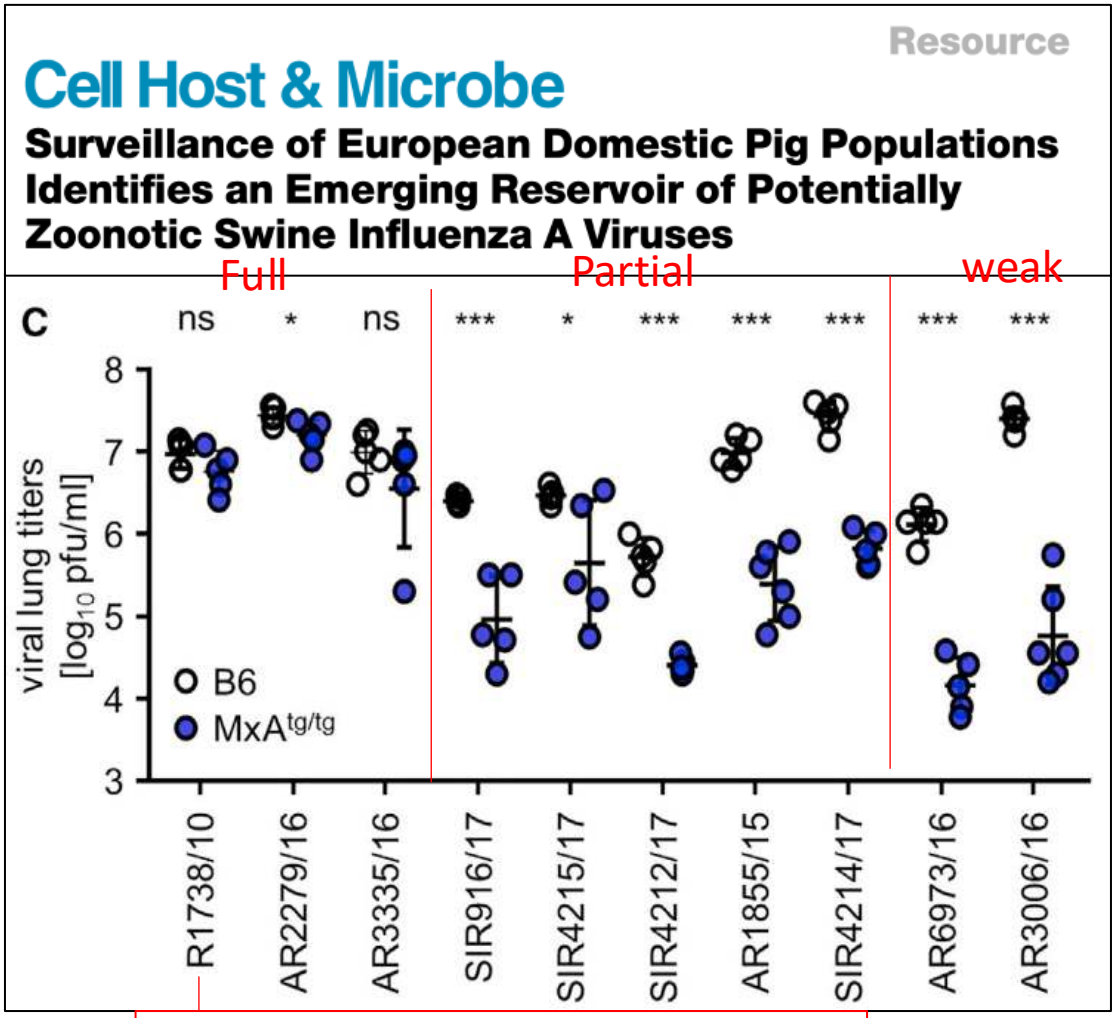


-- Both human and swine respiratory tracks **contain two types of sialic acid receptors but exhibit different distribution.**

Modified from Chauhan1 & Gordon, 2022

The mixing vessel theory

Immune evasion



MxA = Myxovirus Resistance Protein A

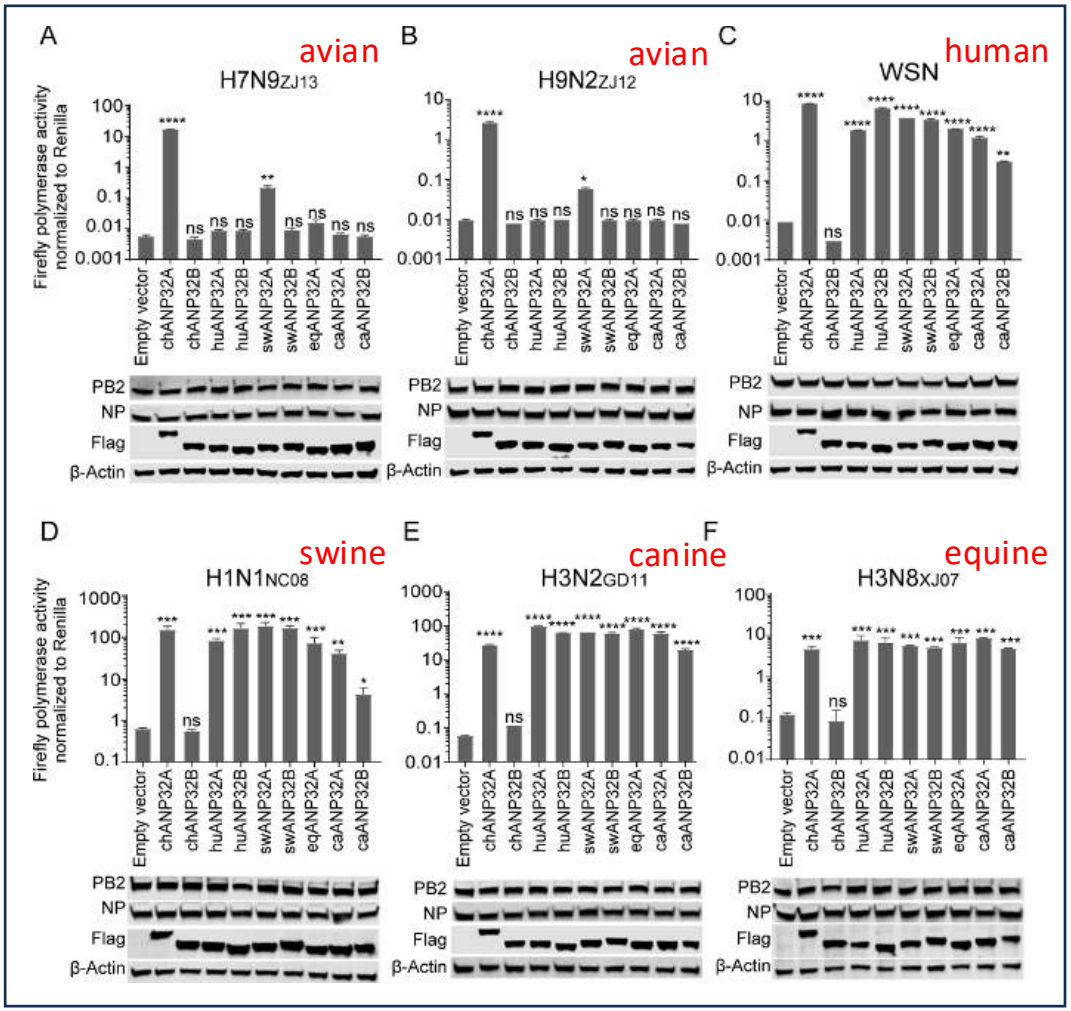
-- Human MxA protein has been shown to be a key inhibitor of IAVs especially of avian origin.

-- Swine Mx1 protein shows less efficient antiviral activity against IAV comparing to human MxA.

-- Currently circulating swine IAVs could possess at least partial MxA resistance that could facilitate human infection with these viruses.

The mixing vessel theory

Replication fitness



Zhang et al, 2020

ANP32 = Acidic leucine-rich nuclear phosphoprotein 32

-- Host ANP32 protein are associated with influenza virus RNA polymerase activity and provide host restriction for mammalian species again avian influenza viruses.

-- Comparing to mammalian ANP32A&B proteins, the swine ANP32A has a stronger ability to support avian influenza virus polymerase.

Epidemiology —Why Swine Influenza matter?

- Pigs can act as reservoirs for older human HA due to frequent reverse zoonosis.
- Selection of human-adapted HA or immune-escape mutants can occur.
- Swine population may serve as building block of IAV with zoonotic or pandemic potential.

Poor Pig Hypothesis

- Pig populations suffer more frequently from reverse zoonotic IAV infections than humans from zoonotic swIAV transmissions
- Differences in **population structures** and **population immunity** of pigs and their keepers
- Co-circulation of an optimally adapted porcine IAV with a newly introduced human IAV would provide reassortment opportunities
- **Personnel in pig farms** should receive **annual vaccinations** against seasonal influenza and staff with respiratory symptoms during the influenza season should avoid contact with pigs.

Transmission

- Direct contact with virus-containing secretion.
- Inhalation of aerosolized viruses

Pathogenesis

- Virus replication is limited to **epithelial cells** of the upper and lower **respiratory tract** of pigs.
- Disease severity of swine IAV infection in pigs depends on the route of infection and viral load in the lower respiratory tract.
- Replication↑ → **proinflammatory cytokines** ↑ → lung lesions ↑
  (high viral load) Double-edged sword → viral clearance
- No virulence determinant has been identified in swine IAV lineages or strains.

Clinical Presentation

Acute course with sudden onset of respiratory illness and rapid recovery.

- Clinical sign:

  -- High fever

  -- Anorexia, inactivity

  -- Coughing, labored breathing, occulonasal discharge

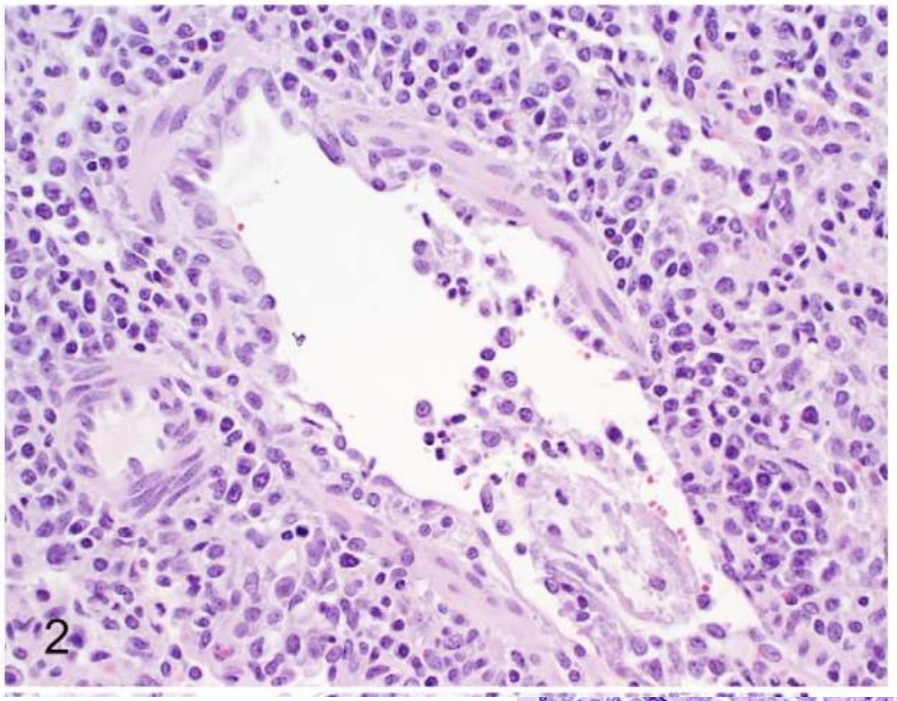
- High morbidity and low mortality in uncomplicated infection
- Concurrent infection or secondary bacterial infection enhance the severity of clinical disease.
- Reduced reproductive performance (ex increased infertility, abortion, small litters, and stillbirths) are occasionally reported.

Lesions



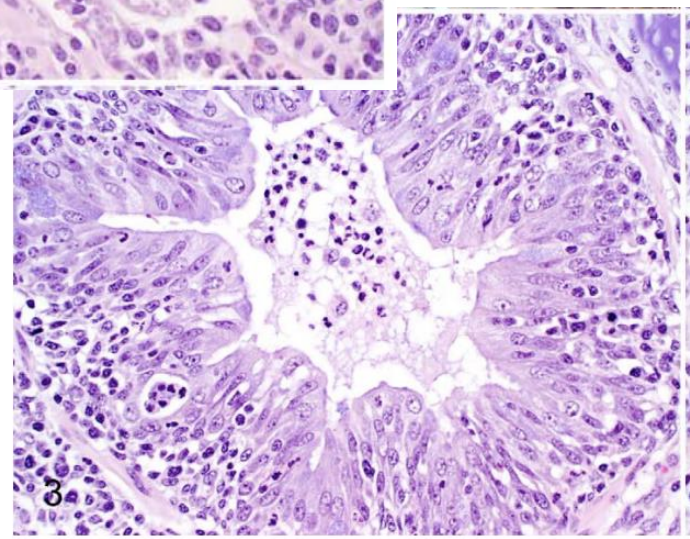
Macroscopic:

The involved lung areas are consolidated, grossly visible and sharply demarcated from the normal tissue; the involved areas will be purple and firm.



Microscopic:

Necrosis of lung epithelia, desquamation/denudation of the bronchial epithelia, and airways obstructed with necrotic epithelial and inflammatory cells, mainly neutrophils



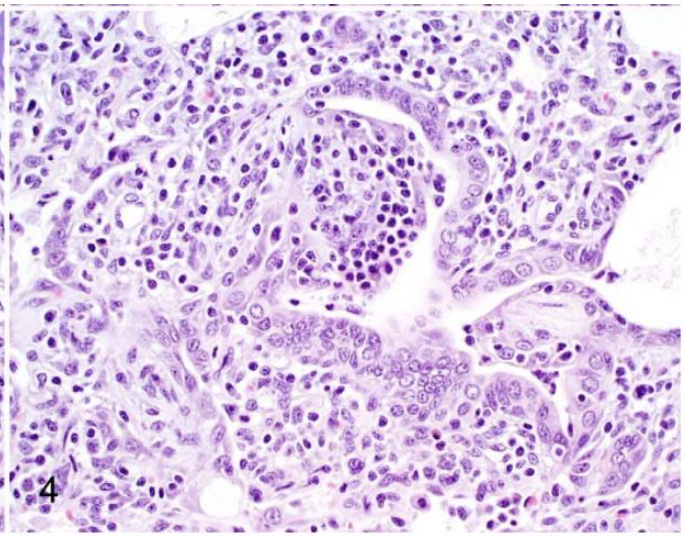


Figure 1. Lung in situ, 4-week-old pig, 5 days after experimental intratracheal inoculation with triple reassortant H3N2 swine influenza virus. Multifocal lobular consolidation in cranial and middle lung lobes is evident. **Figure 2.** Lung, pig, field case. The acute phase of influenza A virus infection is characterized by necrosis and sloughing of bronchiolar epithelial cells and flattening of the remaining cells lining the airway. The lumen contains clusters of neutrophils and sloughed epithelial cells. Light peribronchiolar lymphocyte infiltration is evident. Hematoxylin and eosin (HE). **Figure 3.** Lung, pig, field case. The subacute phase of influenza A virus infection is characterized by an irregular hyperplastic epithelial layer lining the airway. The lumen contains mucus and a few neutrophils and macrophages. Transmigration of the epithelial layer by lymphocytes and a few neutrophil clusters is evident. HE. **Figure 4.** Lung, pig, field case. A small bronchiole is disrupted by an endobronchiolar polyp (bronchiolitis obliterans), a sequel sometimes observed following severe infection with influenza A virus. HE.

Laboratory Diagnosis

There are no pathognomonic signs, and swine influenza must be differentiated from a variety of swine respiratory diseases.

- Objective: confirmation of IAV infection and subtyping
- Virus detection (nasal and pharyngeal swabs during the febrile period of illness)

  -- Identification of (infectious) virion: virus isolation

  **MDCK cells, embryonated eggs

  -- Identification of nucleic acids: RT-PCR, RT-qPCR

Laboratory Diagnosis

- Serological test:
  -- Paired acute and convalescent (3–4-w interval) serum samples
  -- **HI test** (the most common test for specific anti-IAV antibody detection)
  -- ELISA for Ab detection (generic vs subtyping)

Interpretation may be complicated by the co-circulation of different virus subtypes and gene lineages.

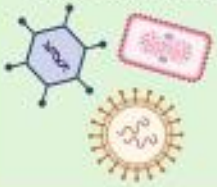
Prevention & Control

Biosecurity and vaccination (Europe and US)

- Most commercial IAV-S vaccines are inactivated whole virus vaccines with an adjuvant. Primary vaccination consists of two intramuscular injections 2–4weeks apart, with twice-yearly booster vaccinations recommended for sows.



https://www.zoetisus.com/products/pork/flusure-xp

Vaccine	Type	Humoral Immunity	Cellular Immunity	MDA Interference	VAERD Induction	Protection	References
Live attenuated influenza virus (LAIV) 	Elastase-dependent	++/-	++	Not tested	Not tested	Heterologous	Masic et al., 2010; Babiuk et al., 2011; Masic et al., 2009; Laudreth et al., 2021; Aubrey et al., 2022
	NS1 truncated and elastase-dependent	+	++	Not tested	Not tested	Homologous and heterologous	Manurow et al., 2019
	NS1 truncated and bat flu vectored	+	++	Not tested	Resistant	Heterologous	Lee et al., 2021
DNA 	Plasmid encoded	++	+	Not tested	Not tested	Homologous	Larsen et al., 2001; Gomes et al., 2011; Bragstad et al., 2013; Borggren et al., 2019; Karlsson et al., 2018
Viral Vectored 	Pichinde Virus (PICV)	+	Not tested	Not tested	Not tested	Homologous	Kumari et al., 2022
	Orf Virus (ORFV)	+++	++	Not tested	Not tested	Homologous	Joshi et al., 2021
	Adenovirus (AdV)	+++	++	Resistant	Resistant	Heterologous	Wesley et al., 2004; Wesley and Lager, 2006; Braucher et al., 2012; Petro-Turnquist et al., 2023
Computationally Designed 	PigMatrix	-	+++	Not tested	Not tested	Homologous	Gutierrez et al., 2015; Gutierrez et al., 2016; Hewitt et al., 2019
	Epigraph	+++	+++	Not tested	Not tested	Heterologous	Bullard et al., 2022
	Consensus (H1 and H3)	H1:+ H3: ++	H1:++ H3: ++	Not tested	Not tested	Heterologous	do Nascimento et al., 2023; Sun et al., 2019
Nanovaccine 	Polyanhydride	+ (+ CpG agonist)	+++	Not tested	Resistant	Heterologous	Dhakal et al., 2017a; Dhakal et al., 2019
	PLGA	+	+++	Not tested	Resistant	Heterologous	Hiremath et al., 2016; Dhakal et al., 2017b
	Chitosan	+++	+++	Resistant (+mannose)	Resistant	Heterologous	Dhakal et al., 2018; Renu et al., 2020; Renu et al., 2021

下列何種病毒特性或結構是造成豬流感病毒容易被消毒劑不活化的原因？

(A) 核衣蛋白
(B) 封套
(C) 分段基因結構
(D) 具重組特性

下列何者不是豬流感病毒在豬隻肺臟常造成的病變？

(A) 肺臟上皮脫落
(B) 支氣管上皮脫落
(C) 支氣管旁腺體壞死和增生
(D) 支氣管阻塞

人、禽、豬的流行性感冒病毒可在豬隻體內重新組合而產生2009年人類新興流感H1N1病毒，其最主要機制 是下列何項？

(A) 豬是所有流感病毒主要自然保毒者（reservoir）
(B) 豬呼吸道細胞同時具有人類與禽類流感病毒HA蛋白接受器
(C) 禽流感病毒在禽類複製偶而產生突變而可以跨物種傳播
(D) 禽流感病毒NS1蛋白可影響豬隻免疫系統，使豬隻容易感染